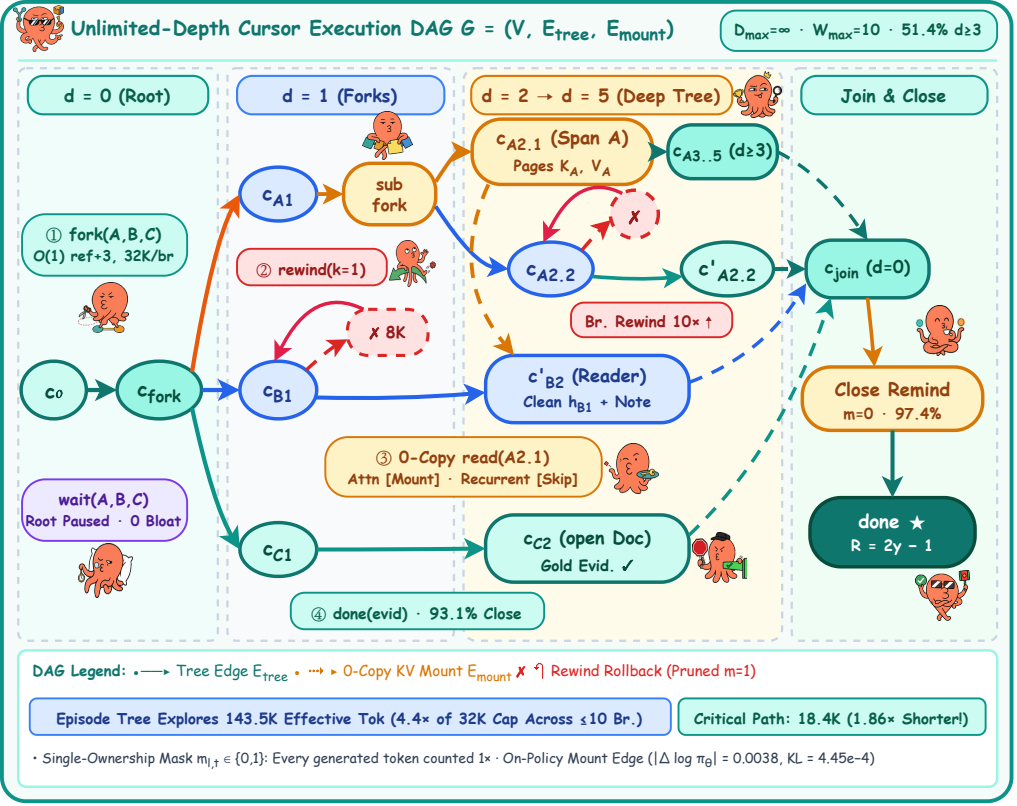

Unlimited-Depth Cursor Execution DAG G = (V, E_tree, E_mount)
D_max=∞ · W_max=10 · 51.4% d≥3
d = 0 (Root)
d = 1 (Forks)
d = 2 → d = 5 (Deep Tree)
Join & Close
① fork(A,B,C)
O(1) ref+3, 32K/br
② rewind(k=1)
✗ 8K
wait(A,B,C)
Root Paused · 0 Bloat
c0
cfork
cA1
sub fork
cB1
cC1
cA2.1 (Span A)
Pages K_A, V_A
cA3..5 (d≥3)
✗
cA2.2
c'A2.2
Br. Rewind 10× ↑
c'B2 (Reader)
Clean h_B1 + Note
③ 0-Copy read(A2.1)
Attn [Mount] · Recurrent [Skip]
cC2 (open Doc)
Gold Evid. ✓
④ done(evid) · 93.1% Close
cjoin (d=0)
Close Remind
m=0 · 97.4%
done ★
R = 2y − 1
DAG Legend: Tree Edge E_tree · 0-Copy KV Mount E_mount ✗ Rewind Rollback (Pruned m=1)
Episode Tree Explores 143.5K Effective Tok (4.4× of 32K Cap Across ≤10 Br.)
Critical Path: 18.4K (1.86× Shorter!)
• Single-Ownership Mask m_i,t ∈ {0,1}: Every generated token counted 1× · On-Policy Mount Edge (|Δ log π_θ| = 0.0038, KL = 4.45e−4)
① Parallel fork & Deep Tree
99.2% Fork
• O(1) Radix Fork: Shares KV prefix (ref+K), clones GDN state h_t, fresh 32K cap/branch.
• Sub-Critical Galton-Watson (μ=0.65 < 1): 5.5 br/ep; 51.4% grow depth d ≥ 3 naturally.
d=1: 48.6%
d=2: 33.8%
d≥3: 51.4% ★
② State Rollback & Close Contract
97.4% done
• Exact State Rollback: Resets cursor c_k, restores GDN h_k & 32K cap (Root ↓ 2.9×, Br. ↑ 10×).
• Close Reminder (m=0) & 2K Reserve: Converts 97.4% roots & 93.1% br. into valid done (0.2% drift).
Root Rw: 29%→10%
Br. Rw: 3.2%→45.1% (10× ↑)
③ Layer-Selective Hybrid KV Mount
21×↓ Hijack
Sibling A KV
Pages K_A, V_A
Reader B State
Private h_B, Δ_B
× 8 Hybrid Blocks in Qwen3.5 (32 Layers)
GDN Layer 4k+1: h_B ← f(h_B, Δ_B) [✗ Skip A]
GDN Layer 4k+2: h_B ← f(h_B, Δ_B) [✗ Skip A]
GDN Layer 4k+3: h_B ← f(h_B, Δ_B) [✗ Skip A]
24 GDN
Full-Attn Layer 4k+4 (ℓ ∈ L_mount): 0-Copy [K_trunk ‖ K_A (0-Copy) ‖ K_B] @ Monotonic s(c'_B)
8 Attn
24 GDN: Private h_B
8 Attn: 97.3% Rank · 2.7% Hijack